CLL — Rai/Binet Staging & Prognostic Risk Map

1. Watch & wait

WHAT YOU SEE

Banner WATCH AND WAIT = STANDARD

WHAT IT ENCODES

Asymptomatic early-stage CLL (Rai 0-II / Binet A) is observed, not treated — 'watch and wait' is the standard of care. Multiple trials (including early FCR studies) confirmed that treating asymptomatic patients does NOT prolong survival and only adds toxicity. Treatment is triggered by active disease, not by counts.

BOARD TRAP

Do not treat asymptomatic CLL just because the WBC/lymphocyte count is high — even very high. Indications to treat (iwCLL): progressive marrow failure (Hb <10 or plt <100), massive/progressive/symptomatic splenomegaly or nodes, lymphocyte doubling time <6 months, autoimmune cytopenia refractory to steroids, or constitutional ('B') symptoms. An absolute lymphocyte count alone is NEVER an indication.

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2. IGHV mutated 60%

WHAT YOU SEE

Gauge MUTATED 60% indolent FCR durable

WHAT IT ENCODES

IGHV-MUTATED CLL (≥2% deviation from germline) is favorable — indolent course, longer time to treatment, and durable remissions. In mutated IGHV without TP53 disruption, FCR produces a plateau on the survival curve suggesting functional cure in a subset. ~50-60% of patients are mutated.

BOARD TRAP

Counterintuitive: IGHV-MUTATED is the GOOD prognosis. Elsewhere in oncology 'mutation = bad', but here the mutated B-cell receptor reflects a post-germinal-center, more differentiated, less proliferative clone.

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3. IGHV unmutated 40%

WHAT YOU SEE

Gauge UNMUTATED 40% aggressive

WHAT IT ENCODES

IGHV-UNMUTATED CLL (<2% from germline, pre-germinal-center) is adverse — faster progression, shorter time to first treatment, and shorter remissions with chemoimmunotherapy. These patients lack the FCR plateau and are preferentially treated with targeted agents.

BOARD TRAP

Unmutated IGHV = aggressive. Do NOT give chemoimmunotherapy (FCR) preferentially here — the benefit/cure plateau is absent; BTKi or venetoclax-based therapy is favored.

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4. TP53 4 pts

WHAT YOU SEE

TP53 = 4 PTS, top of scoring board

WHAT IT ENCODES

In the CLL-IPI, TP53 disruption (del(17p) and/or TP53 mutation) carries the heaviest weight — 4 points, the single most important adverse factor. TP53 sits on chromosome 17p, so del(17p) and TP53 mutation converge on loss of p53 function and chemoresistance.

BOARD TRAP

TP53/del(17p) → AVOID chemoimmunotherapy entirely (chemo works through p53-dependent apoptosis). Use BTK inhibitors or venetoclax. Always test for TP53 mutation AND del(17p) by FISH before EACH line of therapy, as TP53 status can be acquired over time.

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5. IGHV unmut 2 pts

WHAT YOU SEE

Second red row of the right-hand CLL-IPI scoreboard reading IGHV UNMUTATED = 2 PTS — the second-heaviest weight, just below TP53

WHAT IT ENCODES

CLL-IPI assigns 2 points for unmutated IGHV — the second-heaviest single factor after TP53. It captures the biologic aggressiveness independent of stage.

BOARD TRAP

IGHV status is a stable, one-time test (it does not change over the disease course), unlike TP53/FISH which must be repeated before each treatment.

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6. β2M >3.5 2pts

WHAT YOU SEE

Yellow scoreboard row reading β2M >3.5 = 2 PTS — the blood tumor-burden marker scoring as much as IGHV

WHAT IT ENCODES

Beta-2-microglobulin >3.5 mg/L adds 2 points — a serum surrogate for tumor burden and turnover. Elevated β2M also flags reduced renal clearance, so interpret alongside renal function.

BOARD TRAP

β2M is a tumor-burden marker shared with myeloma staging; an isolated elevation reflects bulk/turnover, not a specific genetic lesion.

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7. Rai/Age pts

WHAT YOU SEE

Lower scoreboard rows: RAI I-IV = 1 PT and AGE >65 = 1 PT — clinical/host factors worth only a single point each

WHAT IT ENCODES

CLL-IPI adds 1 point each for advanced clinical stage (Rai I-IV / Binet B-C) and for age >65. These combine clinical and host factors with the biology (TP53, IGHV, β2M) for an integrated score.

BOARD TRAP

Stage and age are only 1 point each — biology (TP53 = 4, IGHV = 2) dominates the CLL-IPI, reflecting that genetics outweigh clinical stage in modern prognosis.

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8. CLL-IPI bands

WHAT YOU SEE
Risk-band table mapping total score to colored survival tiers (low/intermediate/high/very-high) — the score sums the rows above into a prognosis

WHAT IT ENCODES
CLL-IPI total score (0-10) stratifies into four risk bands: low (0-1, 5-yr OS ~93%), intermediate (2-3, ~79%), high (4-6, ~63%), and very high (7-10, ~23%). The score guides prognosis discussions, not the choice of agent directly (targeted therapy is used across all bands).

BOARD TRAP
The CLL-IPI predicts survival but does NOT by itself trigger treatment — asymptomatic high-IPI patients are still watched. Active-disease (iwCLL) criteria, not the IPI score, start therapy.

9. del13q sole BEST

WHAT YOU SEE
del(13q) SOLE = BEST, green field

WHAT IT ENCODES
Isolated del(13q) (loss of miR-15a/16-1) is the MOST FAVORABLE cytogenetic abnormality and the most common, found in ~50%. As a sole abnormality it carries a prognosis similar to or better than normal karyotype, with long survival.

BOARD TRAP
del(13q) ALONE = good prognosis. The catch is 'sole' — del(13q) accompanied by del(17p) or del(11q) is governed by the worse lesion, and a high percentage of del(13q)-positive cells can itself confer slightly worse outcome.

10. Trisomy 12 / normal

WHAT YOU SEE
NORMAL and TRISOMY 12 = intermediate

WHAT IT ENCODES
Normal FISH and trisomy 12 confer intermediate risk. Trisomy 12 is associated with atypical CLL morphology/immunophenotype and sometimes with NOTCH1 mutations.

BOARD TRAP
Trisomy 12 is intermediate — neither the best (del13q) nor worst (del17p). Don't reflexively label any chromosomal GAIN as good or bad without recalling this is the intermediate bucket.

11. del11q ATM adverse

WHAT YOU SEE
del(11q) ATM, volcano, ADVERSE

WHAT IT ENCODES
del(11q) targets the ATM gene (DNA-damage response) and is ADVERSE — classically associated with bulky/massive lymphadenopathy and abdominal disease, younger age, and rapid progression. Historically poor with chemo, it responds well to BTK inhibitors, which substantially improve its outcomes.

BOARD TRAP
del(11q) = bulky nodes + adverse course. The board pivot: although chemo did poorly, BTKi notably overcome the del(11q) disadvantage — so it is no longer the second-worst lesion in the targeted era.

12. del17p WORST

WHAT YOU SEE
del(17p) -70%, WORST red zone

WHAT IT ENCODES
del(17p) (loss of TP53) is the WORST cytogenetic abnormality — chemoresistant, short remissions, shortest survival historically. It is functionally equivalent to TP53 mutation and the two are tested together.

BOARD TRAP
del(17p)/TP53 → NEVER use FCR or any chemoimmunotherapy (p53-independent killing required). Use BTKi or venetoclax. Re-test FISH/TP53 before every line because del(17p) clones can be selected for by prior chemo.

13. Venetoclax road

WHAT YOU SEE
VENETOCLAX path through map

WHAT IT ENCODES
Targeted agents (venetoclax + BTK inhibitors) are effective across ALL cytogenetic and molecular subgroups, including del(17p)/TP53 — the path that crosses the entire risk map. This is why FISH/TP53 status no longer excludes patients from effective therapy, only from chemo.

BOARD TRAP
The presence of del(17p) does NOT mean 'untreatable' — it means 'don't use chemo'; venetoclax and BTKi still work well.

14. Rai ladder 0-IV

WHAT YOU SEE
Ladder: 0 lymphocytosis → IV plt low

WHAT IT ENCODES
Rai staging climbs by disease bulk then marrow failure: 0 = lymphocytosis only (low risk); I = + lymphadenopathy; II = + spleno/hepatomegaly (I-II intermediate); III = + anemia (Hb <11); IV = + thrombocytopenia (plt <100) (III-IV high risk). Binet (A/B/C) parallels this by counting involved node areas and cytopenias.

BOARD TRAP
Stage III/IV cytopenias must result from MARROW INFILTRATION, not autoimmune destruction. AIHA or ITP causing the same Hb/platelet numbers does NOT upstage to Rai III/IV. Also note Rai III is defined by Hb <11 (not <10, which is a treatment trigger) and stage uses platelet <100.